# STACKS, DISCRIMINANTS: LEMMA-LOCALLY-COMES-FROM-UNIVERSAL

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 2224-2242 of sources/stacks-discriminant/pinned-original-source.tex; proof span is lines 2244-2262, the complete author proof (17 body lines).

*Editorial note.* Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Editorial note.* Queue index 40 of queue_discriminant.json (pb=591, stmt 2224-2242, proof 2244-2262).

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

*Source reference.* Original lemma 0FK8: sources/stacks-discriminant/pinned-original-source.tex, source lines 2168–2185. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0BWE: sources/stacks-discriminant/pinned-original-source.tex, source lines 2005–2026. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

**Lemma 0.1.** *Let $f : Y \to X$ be a morphism of schemes. If $f$ satisfies the equivalent conditions of Lemma 0BWE then for every $y \in Y$ there exist $n, d$ and a commutative diagram*

$$\begin{array}{ccccc} Y & \longleftarrow & V & \longrightarrow & Y_{n,d} \\ \downarrow & & \downarrow & & \downarrow \\ X & \longleftarrow & U & \longrightarrow & X_{n,d} \end{array}$$

*where $U \subset X$ and $V \subset Y$ are open with $y \in V$, where $Y_{n,d} \to X_{n,d}$ is as in Example 0FK8, and where the square on the right hand side is cartesian.*

*Proof.* By Lemma 0BWE we can choose $U$ and $V$ affine so that $U = \mathrm{Spec}(R)$ and $V = \mathrm{Spec}(S)$ with $S = R[y_1, \ldots, y_n]/(g_1, \ldots, g_n)$. With notation as in Example 0FK8 if we pick $d$ large enough, then we can write each $g_i$ as $g_i = \sum_{E \in T} g_{i,E} y^E$

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with $g_{i,E} \in R$. Then the map $A \to R$ sending $a_{i,E}$ to $g_{i,E}$ and the map $B \to S$ sending $x_i \to y_i$ give a cocartesian diagram of rings

$$\begin{array}{ccc} S & \longleftarrow & B \\ \uparrow & & \uparrow \\ R & \longleftarrow & A \end{array}$$

which proves the lemma. $\square$